

APPLIED STATISTICS, PROBABILITY THEORY AND MATHEMATICAL STATISTICS

Attila Barta, Sándor Pecsora

Independence of events

Discrete random variables,

Parameters of discrete random variables

Exercise 6.1 (M1)

A coin is tossed ten times. Let A denote the event that there are both heads and tails among the results, while B denotes the event that there is at most one tail among them. Are A and B independent?

A – there is at least a head and at least a tail

B – there is at most one tail

$$P(AB) \stackrel{?}{=} P(A)P(B)$$

$$P(AB) = \frac{10}{2^{10}}$$

$$P(A) = 1 - \frac{2}{2^{10}} = \frac{2^9 - 1}{2^9}$$

$$P(B) = \frac{1 + 10}{2^{10}} = \frac{11}{2^{10}}$$

$$\frac{10}{2^{10}} \neq \frac{11}{2^{10}} \times \frac{2^9 - 1}{2^9}$$

Exercise 6.1 (M1)

MATLAB solution

```
N = 10^7;  
A = 0;  
B = 0;  
AB = 0;  
for i=1:N  
    sim = randi(2,1,10)-1;  
    s = sum(sim);  
    if s>0 & s<10  
        A = A+1;  
    end  
    if s<=1  
        B = B+1;  
    end  
    if s==1  
        AB = AB+1;  
    end  
end  
End  
(A/N) * (B/N)  
AB/N
```

Exercise 6.3

In a box we have 8 cards numbered from 1 to 8. A card is chosen randomly. Let events A, B and C denote the following:

A: the chosen number is even;

B: the chosen number is not greater than 4;

C: the chosen number is either 2 or greater than 5.

Show that $P(ABC) = P(A)P(B)P(C)$ and the three events are not mutually independent.

$$P(A) = \frac{1}{2}$$

$$P(B) = \frac{1}{2}$$

$$P(C) = \frac{1}{2}$$

$$P(ABC) = \frac{1}{8}$$

$$P(AC) \neq P(A)P(C)$$

Exercise 6.7 (M2)

Two soldiers shoot on a target in turn until the first hit. The probability that the starter hits it is 0.2, while the second hits it with probability 0.3. What is the probability that the first successful shot belongs to the soldier who started the shooting?

p – the probability, that the first soldier hits the target first

$P_1(H_i)$ – the probability, that the first soldier hits the target in round i

$P_2(H_i)$ – the probability, that the second soldier hits the target in round i

$$P_1(H_1) = \frac{2}{10}$$

$$P_1(H_2) = \frac{8}{10} \times \frac{7}{10} \times \frac{2}{10}$$

$$p = P_1(H_1) + P_1(H_2) + \dots = \sum_{k=0}^{\infty} 0.2 \times 0.8^k \times 0.7^k = 0.2 \times \sum_{k=0}^{\infty} 0.8^k \times 0.7^k = 0.2 \times \frac{1}{1 - 0.56}$$

Exercise 6.7 (M2)

MATLAB solution

```
f = 0;  
s = 0;  
n = 1000;  
for i=1:n  
    while true  
        if rand()<.2  
            f = f+1;  
            break;  
        elseif rand()<.3  
            s = s+1;  
            break;  
        end  
    end  
end  
f/n
```

0.2 / 0.44

Exercise 7.1

Which of the following sequences form a discrete probability distribution?

a) $p^4, 4p^3q, 6p^2q^2, 4pq^3, q^4$; $q = 1 - p$; $0 < p < 1$;

b) $p^k q^2$; $q = 1 - p$; $0 < p < 1$; $k = 1, 2, \dots$;

a) $(p + q)^4 = (p + (1 - p))^4 = 1$

b) $\sum_{k=1}^{\infty} p^k q^2 = (1 - p)^2 \sum_{k=1}^{\infty} p^k = (1 - p)^2 p \sum_{k=0}^{\infty} p^k = (1 - p)^2 p \frac{1}{1 - p} = (1 - p)p \neq 1$

Exercise 8.1 (M3) Just in Matlab

Two dice are rolled till the first six appears on one of them. What is the mean number of rolls required, inclusive the last one?

ξ – number of rolls till the first 6 appears

$$P(\xi = 1) = \frac{11}{36}$$

$$P(\xi = 2) = \frac{25}{36} * \frac{11}{36}$$

$$P(\xi = 3) = \left(\frac{25}{36}\right)^2 * \frac{11}{36}$$

$$P(\xi = k) = \left(\frac{25}{36}\right)^{k-1} * \frac{11}{36}$$

Geometric distribution:

$$P(\xi = k) = (1 - p)^{k-1} * p$$

$$\mathbb{E}\xi = \sum_{k=1}^{\infty} k * (1 - p)^{k-1} * p = p * \sum_{k=1}^{\infty} k * (1 - p)^{k-1}$$

$$\sum_{k=0}^{\infty} x^k = \frac{1}{1 - x}$$

$$0 < x < 1$$

$$\left(\sum_{k=0}^{\infty} x^k\right)' = \sum_{k=1}^{\infty} k * x^{k-1}$$

$$\left(\frac{1}{1 - x}\right)' = \frac{1}{(1 - x)^2}$$

$$\sum_{k=1}^{\infty} k * x^{k-1} = \frac{1}{(1 - x)^2}$$

$$\mathbb{E}\xi = p \sum_{k=1}^{\infty} (1 - p)^{k-1} = p \frac{1}{(1 - (1 - p))^2} = \frac{1}{p} = \frac{36}{11}$$

Exercise 8.1 (M3)

MATLAB solution

```
n = 10^7;
```

```
s = 0;
```

```
for i=1:n
```

```
    while true
```

```
        s=s+1;
```

```
        if rand()<1/6 | rand()<1/6
```

```
            break;
```

```
        end
```

```
    end
```

```
end
```

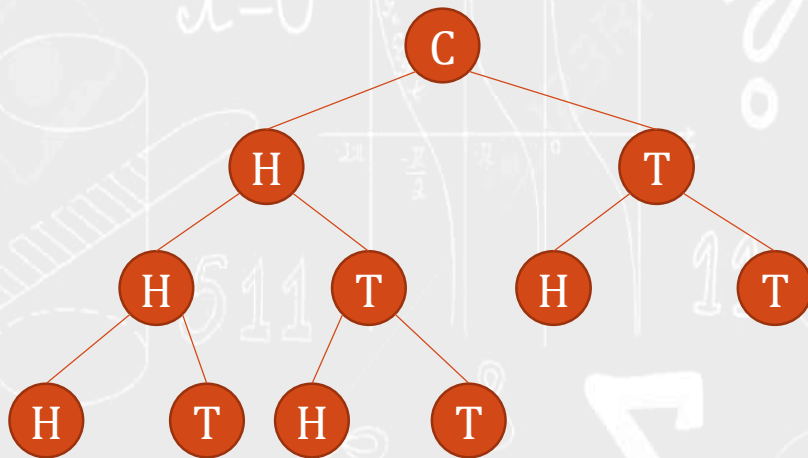
```
s/n
```

```
36/11
```

Exercise 8.4 (M4)

A coin is tossed. If the result is a head, it is tossed two more times, otherwise it is tossed one more time. What is the mean number of heads obtained?

ξ – number of heads



$$P(\xi = 0) = \frac{1}{4}$$

$$P(\xi = 1) = \frac{1}{4} + \frac{1}{8} = \frac{3}{8}$$

$$P(\xi = 2) = \frac{1}{8} + \frac{1}{8} = \frac{2}{8}$$

$$P(\xi = 3) = \frac{1}{8}$$

$$\mathbb{E}\xi = 0 * \frac{1}{4} + 1 * \frac{3}{8} + 2 * \frac{2}{8} + 3 * \frac{1}{8} = \frac{5}{4}$$

Exercise 8.4 (M4)

MATLAB solution

```
n = 10^6;  
s = 0;  
for i=1:n  
    if rand() < 1/2  
        s = s + 1;  
        if rand() < 1/2  
            s = s + 1;  
        end  
        if rand() < 1/2  
            s = s + 1;  
        end  
    else  
        if rand() < 1/2  
            s = s + 1;  
        end  
    end  
end  
s/n  
5/4
```

Exercise 8.7 (M5)

The nominal value of a share is one golden galleon. In a year the value either can be doubled, or halved or remain the same - each of the events has the same probability. On the next year the same happens, independently of the events of the previous year. Find the distribution of the value of the share after two years. What is the mean and the variance of the value?

$$P(\xi = 4) = \frac{1}{9}$$

$$P(\xi = 2) = \frac{2}{9}$$

$$P(\xi = 1) = \frac{3}{9}$$

$$P\left(\xi = \frac{1}{2}\right) = \frac{2}{9}$$

$$P\left(\xi = \frac{1}{4}\right) = \frac{1}{9}$$

$$\mathbb{E}\xi = \frac{49}{36}$$

$$\mathbb{E}\xi^2 = 16 * \frac{1}{9} + 4 * \frac{2}{9} + 1 * \frac{3}{9} + \frac{1}{4} * \frac{2}{9} + \frac{1}{16} * \frac{1}{9} = \frac{49}{16}$$

$$\mathbb{D}\xi^2 = \mathbb{E}\xi^2 - (\mathbb{E}\xi)^2 = \frac{98}{81}$$

Exercise 8.7 (M5)

MATLAB solution

```
n = 10^7;  
x = 0;  
xs = 0;  
for i=1:n  
    g = 1;  
    for j=1:2  
        r = rand();  
        if r<1/3  
            g = g*2;  
        elseif r>2/3  
            g = g/2;  
        end  
    end  
    x = x+g;  
    xs = xs+g^2;  
end  
x_mean = x/n  
x_var = xs/n - x_mean^2
```

Literature

- ❑ Ágnes Baran: Mathematics for engineers 1. (Laboratory slides)
- ❑ Sándor Baran: Probability theory and statistics
- ❑ Fazekas István: Valószínűségszámítás, Debreceni Egyetemi Kiadó, 2009
- ❑ Fazekas István: Bevezetés a matematikai statisztikába, Debreceni Egyetemi Kiadó, 2009
- ❑ Matlab examples:
<https://www.mathworks.com/help/examples.html>